

A 25-year-old woman who is 30 weeks pregnant has been referred to your tertiary high-risk obstetric clinic. She has complex cyanotic congenital heart disease and now functions with a Fontan circulation. How would you stratify the cardiovascular risk? What are the issues relevant to anaesthetic care that will need to be managed for this patient?

(From ANZCA 2014B)

Examiner Report

31.9% of candidates passed this question.

As a minimum in

Part A candidates needed to demonstrate an understanding that patient is at increased peripartum CVS risk; and mention at least 2 of the main risk factors

i.e. poor maternal functional class (NYHA Class III or IV) **or** cyanosis; myocardial dysfunction; L heart obstruction; prior arrhythmia or prior cardiac events

In part B relevant issues included ; some attempt to name some likely CVS complications (including death - heart failure should be mentioned]; that the greatest risk was during the peripartum period: good consultation with other specialties including at least cardiology (multidisciplinary team); recognising the need for additional monitoring and observation during the antenatal period and extra resource for delivery e.g. HDU; and that the delivery should be planned with good communication to all parties.

Sample Answer

Answer provided here is aspirational, and not meant to be reproduced under exam conditions. In-depth knowledge of subspecialisation (OBAN-Cardiac) is not expected for registrar level exam.

I would risk stratify patient by based on the **modified WHO classification** and **CARPREG II scoring**. I would review patient's history, physical examination, and investigations specifically looking out for complications associated with Fontan circulation which will place patient at mWHO IV, as opposed to mWHO III (well parturient post-Fontan with no complication).

Factors that increases cardiovascular risk for anaesthesia for this patient:

- History:
 - Poor effort tolerance (NYHA III-IV).
 - Cardiac symptoms (chest pain, SOB, PND, palpitations).
 - Presence of refractory arrhythmias.
- Physical Examination:
 - Poor SpO₂ (<90%) on room air. ○ Presence of murmurs.
 - Bibasal crepitations. Palpable/loud P2. Raised JVP/peripheral oedema.
- Investigations:
 - TTE showing ventricular function (poor EF), valvular abnormalities (in particular, moderate/severe AV regurgitations) pulmonary hypertension, LVOT obstruction.
 - Evidence of protein-losing enteropathy.

This is a patient in third trimester with a complex cardiac condition and univentricular physiology (Fontan), with high cardiovascular risk for anaesthesia.

She should be managed in a tertiary hospital with access to a multidisciplinary team of experienced obstetricians, neonatologists, cardiologists, intensivists, and anaesthetists.

Issues relevant to anaesthetic care that will need to be managed:

Antepartum

- Maternal cardiovascular complications
 - Ideally, prior to her pregnancy, she should have already consulted a cardiologist and obstetrician for (i) optimisation of her heart condition and (ii) counselling on suitability for pregnancy/ labour.
 - Physiological changes of pregnancy (especially CVS and respi) can worsen her cardiac function, and patient should be followed up monthly with her cardiologist.
 - New symptoms such as disproportionate/ unexplained dyspnoea or new murmur should be investigated with TTE.
 - Any need for cardiac intervention (percutaneous or surgical) should be discussed with a multidisciplinary team.
 - Early delivery (foetus is viable at 30 weeks GA) if patient's cardiac status deteriorates.
 - Risk of peri-partum cardiac failure, including death, should be communicated and counselled to patient.
- Maternal obstetric complications
 - Increased risk of pre-eclampsia and gestational DM compared to general population. They should be screened and treated.
 - Increased risk of thrombo-embolic complication, including intra-cardiac thrombus. Patient may need to be started on anti-coagulants, which should be balanced with bleeding risk and possibly central neuraxial use.
- Foetal complications
 - Teratogenicity risk especially with Warfarin during 1st trimester.
 - Foetus may be at higher risk of congenital heart disease and screening should be offered. If foetal congenital heart disease is detected, delivery should be at an institution that can provide neonatal cardiac care.
 - Higher risk of foetal growth restriction. Delaying delivery date may improve foetal survival rate.
- Delivery date (should be planned and discussed during antenatal visit)
 - Induction of labour is recommended at 40 weeks of gestation in women with cardiac disease, which reduces risk of Em LSCS and still births.
 - The duration of gestation should be balanced between adequate intra-uterine growth vs increasing cardiovascular risks to mother during pregnancy.
- Mode of delivery (should be planned and discussed during antenatal visit)
 - Vaginal delivery is recommended as first choice unless contraindicated.
 - Contingency planning, including Em LSCS if maternal cardiac status deteriorates, should be discussed with patient.

Intra-partum

- Antibiotic prophylaxis for IE.
- Anaesthesia/analgesia
 - Choice of CNB (epidural/spinal) vs GA should be discussed.
 - Anti-coagulant (if started) stop and (re)start date will need to be planned to facilitate CNB.
 - Patient is also at risk of pre-term delivery. Hence, patient should also be counselled for GA in the event of unplanned early labour requiring LSCS.
- Haemodynamic monitoring during delivery.
 - Continuous vitals monitoring, including telemetry, in delivery suite.
 - IA line and CVP if patient undergoes GA.

- Having a 2nd (cardiac) anaesthetist who is proficient in TEE will be useful for GA for LSCS.
- Optimisation of physiology in Fontan circulation during labour.
 - Fluid shifts/ autotransfusion and sympathetic stress from contractions during labour can worsen cardiac function, and should be anticipated and managed promptly.
 - Supplemental O₂ to optimise oxygen delivery to minimise myocardial depression ○
 - Maintain normal preload
 - Difficult to assess fluid status clinically as patient may be oedematous.
 - IV fluids should be given judiciously to maintain CO (pre-load dependent). Avoid fluid overload, which can tip patient's solitary ventricle into failure.
 - Flotrac may be used to guide fluid management. Urinary catheter for urine output monitoring.
 - Aggressive treatment of arrhythmias
 - Fluid shifts during labour can to stretching of atrial wall and trigger arrhythmias.
 - Atrial arrhythmias should be promptly treated promptly
 - Electrolytes should be corrected and patient should be kept normothermia.
 - Maintain normal SVR
 - Epidural analgesia should be slowly titrated to relief, to minimise hypotension associated with systemic vasodilation.
 - Vasopressors (phenylephrine) may be needed.
 - Avoid worsening pulmonary hypertension
 - Achieved by minimising pain, hypoxemia, hypercardia, and acidosis.
 - Avoid air bubbles in IV infusion lines
 - Especially if patient has residual left-to-right shunting (PDA/ASD/VSD)

Post-partum

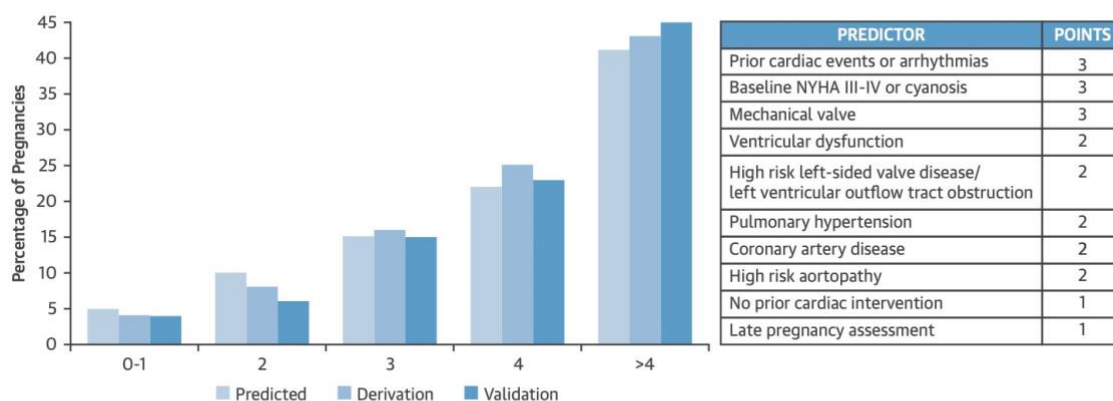
- Disposition post-delivery
 - Patient should be monitored closely during the first 24-48h post-partum in the delivery suite or high dependency unit with continuous telemetry monitoring.
 - Early cardiology review to assess and optimise maternal cardiac status.
- Elevated PPH risk
 - Slow IV infusion of Oxytocin may be administered for uterine contraction. Avoid bolus.
 - Avoid Ergometrine (risk of worsening PVR)
- Elevated thrombo-embolic risk
 - Mechanical thrombo-embolic prophylaxis (compression stockings and calf pumps) should continue during and after labour, especially upon cessation of anti-coagulant.
 - Encourage early ambulation if patient is haemodynamically stable

Table 3 Modified World Health Organization classification of maternal cardiovascular risk

	mWHO I	mWHO II	mWHO II–III	mWHO III	mWHO IV
Diagnosis (if otherwise well and uncomplicated)	Small or mild – pulmonary stenosis – patent ductus arteriosus – mitral valve prolapse Successfully repaired simple lesions (atrial or ventricular septal defect, patent ductus arteriosus, anomalous pulmonary venous drainage) Atrial or ventricular ectopic beats, isolated	Unoperated atrial or ventricular septal defect Repaired tetralogy of Fallot Most arrhythmias (supraventricular arrhythmias) Turner syndrome without aortic dilatation	Mild left ventricular impairment (EF >45%) Hypertrophic cardiomyopathy Native or tissue valve disease not considered WHO I or IV (mild mitral stenosis, moderate aortic stenosis) Marfan or other HTAD syndrome without aortic dilatation Aorta <45 mm in bicuspid aortic valve pathology Repaired coarctation Atrioventricular septal defect	Moderate left ventricular impairment (EF 30–45%) Previous peripartum cardiomyopathy without any residual left ventricular impairment Mechanical valve Systemic right ventricle with good or mildly decreased ventricular function Fontan circulation. If otherwise the patient is well and the cardiac condition uncomplicated Unrepaired cyanotic heart disease Other complex heart disease Moderate mitral stenosis Severe asymptomatic aortic stenosis Moderate aortic dilatation (40–45 mm in Marfan syndrome or other HTAD; 45–50 mm in bicuspid aortic valve, Turner syndrome ASI 20–25 mm/m ² , tetralogy of Fallot <50 mm) Ventricular tachycardia	Pulmonary arterial hypertension Severe systemic ventricular dysfunction (EF <30% or NYHA class III–IV) Previous peripartum cardiomyopathy with any residual left ventricular impairment Severe mitral stenosis Severe symptomatic aortic stenosis Systemic right ventricle with moderate or severely decreased ventricular function Severe aortic dilatation (>45 mm in Marfan syndrome or other HTAD, >50 mm in bicuspid aortic valve, Turner syndrome ASI >25 mm/m ² , tetralogy of Fallot >50 mm) Vascular Ehlers–Danlos Severe (re)coarctation Fontan with any complication
Risk	No detectable increased risk of maternal mortality and no/mild increased risk in morbidity	Small increased risk of maternal mortality or moderate increase in morbidity	Intermediate increased risk of maternal mortality or moderate to severe increase in morbidity	Significantly increased risk of maternal mortality or severe morbidity	Extremely high risk of maternal mortality or severe morbidity
Maternal cardiac event rate	2.5–5%	5.7–10.5%	10–19%	19–27%	40–100%
Counselling	Yes	Yes	Yes	Yes: expert counselling required	Yes: pregnancy contraindicated: if pregnancy occurs, termination should be discussed
Care during pregnancy	Local hospital	Local hospital	Referral hospital	Expert centre for pregnancy and cardiac disease	Expert centre for pregnancy and cardiac disease
Minimal follow-up visits during pregnancy	Once or twice	Once per trimester	Bimonthly	Monthly or bimonthly	Monthly
Location of delivery	Local hospital	Local hospital	Referral hospital	Expert centre for pregnancy and cardiac disease	Expert centre for pregnancy and cardiac disease

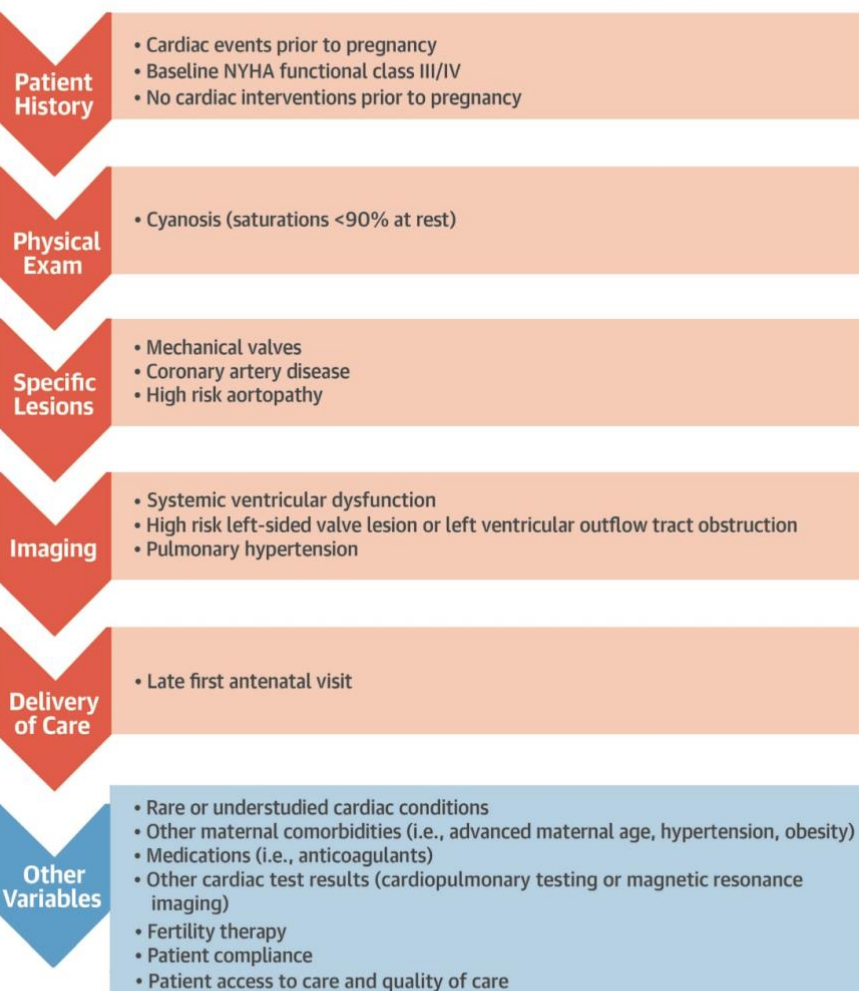
ASI = aortic size index; EF = ejection fraction; HTAD = heritable thoracic aortic disease; mWHO = modified World Health Organization classification; NYHA = New York Heart Association; WHO = World Health Organization.

FIGURE 4 CARPREG II Risk Prediction Index: Incidence of Adverse Cardiac Events Stratified According to CARPREG II Risk Scores



The CARPREG (Cardiac Disease in Pregnancy Study) II risk score is based on 10 predictors, shown in the **box**. Each predictor is assigned a weighted point score. The sum of points represents the risk score. Risk scores are categorized into the 5 groups (x-axis). The predicted (**light blue**) and the observed frequency of primary cardiac events in the derivation (**medium blue**) and validation (**dark blue**) groups are shown on the y axis. NYHA = New York Heart Association.

CENTRAL ILLUSTRATION Predictors of Adverse Events in Pregnant Women With Heart Disease



Silversides, C.K. et al. J Am Coll Cardiol. 2018;71(21):2419-30.

Determining cardiac risk in pregnant women with heart disease requires integration of risk score estimates, individual factors, and clinical judgment. The **red arrows** show the variables in the CARPREG II risk score used to predict adverse cardiac events in pregnant women with heart disease. In addition to the variables in the CARPREG II risk score, there may be other factors that impact outcomes for the individual patient. The **blue arrow** shows some of the other variables to consider when estimating pregnancy risks.

